THE LOWER SHRIEK FUNCTORS ALONG THE THREE PROJECTIONS OF THE STANDARD TRIANGLE ARE INVERSE TO THE PULLBACKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is an explicit three-case analysis; the author marks the three cases with the words 'Proof for i = 1', 'The case i = 2' and 'The case i = 3', all inside the counted proof span.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), sliced mechanically from the chapter preamble.

Allowed standard prerequisites (editorial).

- Computation of left derived lower shriek for a fibred category. Cohomology on Sites, Lemma sites-cohomology-lemma-compute-left-derived-pi-shriek, used at proof lines 1529-1534 and 1553-1558. Gives the description of the derived functors on the fibre categories for i = 1, 2.
- The lower-shriek computation lemma with its fibre categories I_U. Cohomology on Sites, Lemma sites-cohomology-lemma-compute-left-derived-g-shriek, used at proof lines 1564-1582. Reduces the case i = 3 to the vanishing statement on the fibres.
- Vanishing of the higher cohomology of the relevant sheaves on the fibres. Lemma lemma-polynomial-on-top (1438-1444), used at proof lines 1577-1579. Supplies H_n = 0 for n > 0 and the degree zero value in the case i = 3.
- Constant sheaves and initial or final objects in the fibre categories. Lemma lemma-pi-lower-shriek-constant-sheaf and Cohomology on Sites, Lemma sites-cohomology-lemma-initial-final, used at proof lines 1535-1537 and 1559-1561. Settle the cases i = 1 and i = 2.

Source reference. Original equation 08QT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11071–11071. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11057–11057. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08QV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11073–11073. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08PK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11028–11028. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08PJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11027–11027. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08Q7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11050–11050. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *With notation as in (08QT) we have $Lg_{i,!} \circ g_i^{-1} = id$ for $i = 1, 2, 3$ and hence also $L\pi_! \circ g_i^{-1} = L\pi_{i,!}$ for $i = 1, 2, 3$.*

Proof. Proof for $i = 1$. We claim the functor $\mathcal{C}_{C/B/A}$ is a fibred category over $\mathcal{C}_{B/A}$ Namely, suppose given $(P \to B, Q \to C)$ and a morphism $(P' \to B) \to (P \to B)$ of $\mathcal{C}_{B/A}$. Recall that this means we have an A-algebra homomorphism $P \to P'$ compatible with maps to B. Then we set $Q' = Q \otimes_P P'$ with induced map to C and the morphism

$$(P' \to B, Q' \to C) \longrightarrow (P \to B, Q \to C)$$

in $\mathcal{C}_{C/B/A}$ (note reversal arrows again) is strongly cartesian in $\mathcal{C}_{C/B/A}$ over $\mathcal{C}_{B/A}$. Moreover, observe that the fibre category of u_1 over $P \to B$ is the category $\mathcal{C}_{C/P}$. Let $\mathcal{F}$ be an abelian sheaf on $\mathcal{C}_{B/A}$. Since we have a fibred category we may apply Cohomology on Sites, Lemma 08PJ. Thus $L_n g_{1,!} g_1^{-1}\mathcal{F}$ is the (pre)sheaf which assigns to $U \in \mathrm{Ob}(\mathcal{C}_{B/A})$ the nth homology of $g_1^{-1}\mathcal{F}$ restricted to the fibre category over U. Since these restrictions are constant the desired result follows from Lemma 08QE via our identifications of fibre categories above.

The case $i = 2$. We claim $\mathcal{C}_{C/B/A}$ is a fibred category over $\mathcal{C}_{C/A}$ is a fibred category. Namely, suppose given $(P \to B, Q \to C)$ and a morphism $(Q' \to C) \to (Q \to C)$ of $\mathcal{C}_{C/A}$. Recall that this means we have a B-algebra homomorphism $Q \to Q'$ compatible with maps to C. Then

$$(P \to B, Q' \to C) \longrightarrow (P \to B, Q \to C)$$

is strongly cartesian in $\mathcal{C}_{C/B/A}$ over $\mathcal{C}_{C/A}$. Note that the fibre category of u_2 over $Q \to C$ has an final (beware reversal arrows) object, namely, $(A \to B, Q \to C)$. Let $\mathcal{F}$ be an abelian sheaf on $\mathcal{C}_{C/A}$. Since we have a fibred category we may apply Cohomology on Sites, Lemma 08PJ. Thus $L_n g_{2,!} g_2^{-1}\mathcal{F}$ is the (pre)sheaf which assigns to $U \in \mathrm{Ob}(\mathcal{C}_{C/A})$ the nth homology of $g_1^{-1}\mathcal{F}$ restricted to the fibre category over U. Since these restrictions are constant the desired result follows from Cohomology on Sites, Lemma 08Q7 because the fibre categories all have final objects.

The case $i = 3$. In this case we will apply Cohomology on Sites, Lemma 08PK to $u = u_3 : \mathcal{C}_{C/B/A} \to \mathcal{C}_{C/B}$ and $\mathcal{F}' = g_3^{-1}\mathcal{F}$ for some abelian sheaf $\mathcal{F}$ on $\mathcal{C}_{C/B}$. Suppose $U = (\overline{Q} \to C)$ is an object of $\mathcal{C}_{C/B}$. Then $\mathcal{I}_U = \mathcal{C}_{\overline{Q}/B/A}$ (again beware of reversal of

arrows). The sheaf $\mathcal{F}'_U$ is given by the rule $(P \to B, Q \to \overline{Q}) \mapsto \mathcal{F}(Q \otimes_P B \to C)$. In other words, this sheaf is the pullback of a sheaf on $\mathcal{C}_{\overline{Q}/C}$ via the morphism $Sh(\mathcal{C}_{\overline{Q}/B/A}) \to Sh(\mathcal{C}_{\overline{Q}/B})$. Thus Lemma 08QV shows that $H_n(\mathcal{I}_U, \mathcal{F}'_U) = 0$ for $n > 0$ and equal to $\mathcal{F}(\overline{Q} \to C)$ for $n = 0$. The aforementioned Cohomology on Sites, Lemma 08PK implies that $Lg_{3,!}(g_3^{-1}\mathcal{F}) = \mathcal{F}$ and the proof is done. $\square$